

Secure Settings Identify Options Procedure

Revision

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SME

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Abstract

This document describes the procedure used to perform the security option identification activity described in the **AVCDL** ^[1] secondary document **Secure Settings Document** ^[2].

Group / Owner

Security / Security Architect

Motivation

This document is motivated by the need to ensure that the software is built with tools whose selected settings are based on cybersecurity best practices. This is necessary given the nature of safety-critical, cyber-physical systems

Note: Within the context of this document, the terms **security** and **cybersecurity** are used interchangeably. It is presumed that the term **security** is being used in reference to **cybersecurity** and not **physical security**.

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Audience

The audience of this document is the cybersecurity practitioner who will be conducting cybersecurity-specific tool setting options identification.

Completeness of Output

Since the identification of security options is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. As with any cybersecurity activity, it is not the place of the cybersecurity SME to make unsupported assertions. All information should be based on the best available sources and confirmed where possible prior to inclusion.

Note: When information is not available for either a given section of the template or parts thereof, this should be noted.

Disposition of Output

Once completed, the generated output should be managed in the organization's document management system (DMS) as a document of record.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity-relevant tool settings. Further, that the reader has read and understood the AVCDL **Secure Settings Document** secondary document.

Prerequisites – Security SME

Qualifications

It is required that the cybersecurity SME is both a qualified and trained security architect (shown above on title page as **Owner**) as defined by the **NIST NCWF** role SP-ARC-002 and detailed in section **12.7 Security Architect** of the AVCDL primary document [\[1\]](#).

Knowledge

It is required that the cybersecurity SME understands the purpose of cybersecurity-relevant tool settings.

Background Information

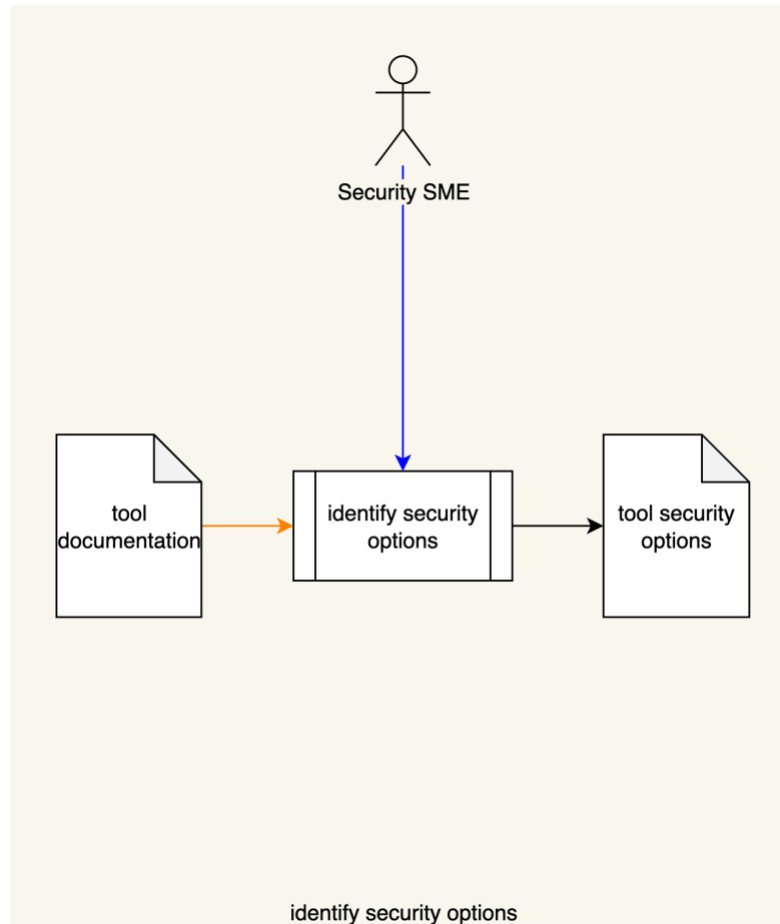
It is required that the cybersecurity SME has read and understands the AVCDL **Secure Settings Document** [\[2\]](#) secondary document. Additionally, that the cybersecurity SME has taken training relevant to this activity.

Prerequisites – Input Materials

It is required that the development group provide all relevant documentation related to the tool under consideration necessary to complete the procedure.

Identify Security Options Activity

The workflow diagram for the **identify security options** activity of the **Secure Settings Document** is shown below.



The Security SME reviews the **Tool Documentation** and identifies the cybersecurity options. The identified options are put in a **Tool Security Options** list.

The **Tool Documentation** may come from numerous sources including, but not limited to:

- Hardcopy
- Online (vendor site)
- Source code

Each entry in the **Tool Security Options** list is minimally composed of the following information:

- Option
- Description of the option
- Security relevance
- Bug Class

The set of **bug class** values is taken from the **NIST Bugs Framework** [\[4\]](#) document.

Note: The list may contain additional information as appropriate to the type of tool.

NIST Bug Classes

The **NIST Bugs Framework** provides a failure taxonomy which is useful in determining which cybersecurity-relevant options to focus on. This information will be used in later activities within the secure setting process.

The bug classes are:

Input

- Validation
- Verification

Memory

- Addressing
- Management
- Use

Data

- Declaration
- Name Resolution
- Type Computation
- Type Conversion

Note: It is assumed that not all tools will have cybersecurity-relevant options covering all bug classes.

Methodology

The methodology for this activity is straightforward.

All available documentation is reviewed and a list of options is assembled.

Note: It is important to include all of the tool's cybersecurity-relevant options, even those which may not be applicable to specific products. Later activities will tailor the list to those which are deemed appropriate.

Tool Security Options Template

The tool's cybersecurity-relevant options may be documented using the **AVCDL tool cybersecurity options template** Microsoft Excel workbook [\[3\]](#).

Note: Other forms of documentation are permissible so long as they provide the information laid out in this document.

There are five sheets in the workbook. They are:

- [Cover sheet](#)
- [Revision history](#)
- [Reference documents](#)
- [Options](#)
- [Legend](#)

These will be addressed in turn.

Duplication of Rows in the Various Sheets

When there is the need to add rows to the various sheets of the workbook, be sure to duplicate an existing row. This is because validation checks are attached to some of the cells which also enables the use of dropdown lists.

Cover Sheet

The **cover sheet** of the workbook is shown below:

Tool Cybersecurity Options									
Tool Name	Tool Name								
Tool Version	Tool Version								
Tool Platform	Tool Platform								
Tool URL	Tool URL								
Cybersecurity SME	Cybersecurity SME								
Date	27-Aug-2000								
Revision	1								

Fields to be completed are shown in **red**.

Tool Name

This is the name of the tool whose cybersecurity options are being identified.

Tool Version

This is version number of the tool under consideration.

Tool Platform

This is the platform on which the tool is run.

Tool URL

This is location where the tool can be found.

Cybersecurity SME

This is the cybersecurity subject matter expert creating the options list.

Date

This is the date when the options list was created or updated. The date should be updated whenever the list is updated.

Revision

This is the revision number of the document. The revision number is a monotonic and increasing integer, starting at 1. It should be incremented every time the document is updated.

Revision History

The **revision history** sheet of the workbook is shown below:

Revision History		
Revision	Author	Description
1		initial revision

Revision

The **revision** corresponds to that listed on the cover sheet.

Author

The **author** corresponds to the cybersecurity SME listed on the cover sheet.

Description

This is a brief description of changes made to the workbook since it was last updated.

Reference Documents

Note: These references are those necessary for the creation of the cybersecurity options list.

The **references** sheet of the workbook is shown below:

Reference Documents		
Name	Description	Location

Name

This is the name of the document being referenced.

Description

This is a brief description of the document being referenced.

Note: In cases where the reference material is expressed in the form other than a document (for example, a web page), this should be noted.

Location

This is the location of the document being referenced. It may be a physical location or a URL.

Options

The **options** sheet of the workbook is shown below:

Options					
ID	option	description	relevance	bug class	notes

ID

This is the unique ID of the option.

Note: An ID of the form `<TOOL>-<zero-leading number>` is recommended. Where multiple tools of the same type (such as compilers) use the same options, a generalized tool type should be used for the ID's prefix.

Option

This is the name of the option.

Description

This is the description of the option.

Relevance

This field describes why the options is relevant from a cybersecurity perspective.

Bug Class

This is the NIST bug class of the option.

Notes

This is a general notes field.

Legend

The **legend** sheet of the workbook is shown below:

NIST bug class
input validation
input verification
memory addressing
memory management
memory use
data declaration
data name resolution
data type computation
data type conversion

The **legend** sheet information is used to make the completion of the document easier by providing dropdown lists for common values. It also ensures that spelling errors do not creep into the generated material.

Note: The legend sheet should not be edited. If an unlisted value is required, the template should be separately revised.

Exit Criteria

This procedure is considered complete once the generated output has been entered into the organization's DMS as a document of record.

Note: The processes and procedures for entering documents into the DMS, or the updating thereof, are outside the scope of this document.

References

1. **AVCDL** (AVCDL primary document)
2. **Secure Settings Document** (AVCDL secondary document)
3. **AVCDL tool cybersecurity options template** (AVCDL template)
4. **NIST SP 800-231 Bugs Framework (BF): Formalizing Cybersecurity Weaknesses and Vulnerabilities**
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-231.pdf>